

Predicate Logic

What is predicate logic?

- Consider the following sentence:

Every person who is 18 years or older, is eligible to vote

- Can express the sentence using propositional logic only?
- The answer is: No
 - In the example sentence, there are two variables: person and age
 - The person is depended on age
 - 'Eligible to vote?' – can be answered using true or false.
 - That means this part is a simple logic
 - However, 'Every person who is 18 years or older' depends specific condition.
 - We cannot express it using simple logic
- That means we need a more powerful logic to express it.
- And that is predicate logic.

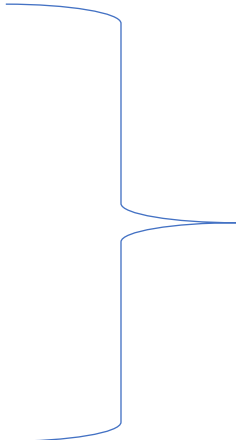
What is predicate logic?

- Predicate logic is an extension of propositional logic.
- It combines the concept of predicate and quantifiers to better capture the meaning of statements that cannot be meaningfully expressed by propositional logic only.

What is predicate?

- Consider the statement: '5 is greater 3'
 - Is it a statement?
 - Yes: '5 is greater 3'
- Consider another statement: 'X is greater than 3'
 - Is it a statement?
 - The answer is not straightforward.
 - Consider the following scenario:

Value of X	Value of the Statement
X = 1	False
X = 3	False
X = 5	True



A proposition can have either true or false value. And they cannot interchange

What is predicate?

- The sentence: 'X is greater than 3'
- Let's dissolve the sentence into two parts: Subject and predicate
 - Subject: x
 - Predicate: 'is greater than 3'
- That means, predicate is the part of the sentence that does not change.
- The value of the predicate depends on subject
- Here, the value (true or false) of 'is greater than 3' depends on x
- Therefore, the sentence is a function of x

What is predicate?

- We can express the sentence 'X is greater than 3' using a mathematical function $P(x)$
- Here, P is the predicate and x is the subject (or variable)
- Now $P(x)$ can be either true or false depending on the value of x .
- Once the value of x is assigned, then $P(x)$ can be either true or false but not both at the same time.
- And for that particular x , it never changes.
- That means $P(x)$ becomes a proposition.

Example of Predicate Logic

- Example 1: $P(x): x > 10$. What are the truth values of $P(15)$ and $P(5)$?
 - $P(15) \cong \text{True}$
 - $P(5) \cong \text{False}$
- Example 2: $R(x, y): x = y + 1$. What is the truth value of the proposition $R(1, 3)$ and $R(2, 1)$?
 - $R(1, 3): 1 = 3 + 1 \cong \text{False}$
 - $R(2, 1): 2 = 1 + 1 \cong \text{True}$